Competency	Competency Level	Contents	Learning outcomes	Periods
			control structures in programs	
	9.9 Uses sub-programs in programming	• Types of subprograms ○ Built in ○ User defined ❖ Structure ❖ Parameter passing ❖ Return values ❖ Default values ❖ Scope of variables	• Briefly describes the functions • Lists and briefly describes the types of functions • Identifies the structure of a function • Compares local and global variables • Identifies the behavior of a variable in terms of life time • Identifies the need of return values and writes functions to obtain the appropriate return value • Writes functions using relevant parameters and arguments • Uses user defined functions	10
	9.10 Uses data structures in programs	• Data structures ○ Strings ○ Lists	• Briefly explains the use of data structures	8

Competency	Competency Level	Contents	Learning outcomes	Periods
		○ Tuples ○ Dictionaries	• Uses relevant data structures in programming	
	9.11 Handles files and databases in programs	• File handling ○ Basic file operations	• Uses basic file operations (open, close, read write and append)	6
	9.12 Manages data in databases	• Connecting to a database • Retrieve data • Add, modify and delete data	• Embeds SQL statements in programming languages to retrieve, add, modify and delete data	4
	9.13 Searches and sorts data	• Searching techniques ○ Sequential search • Sorting techniques ○ Bubble sort	• Uses sequential searching technique appropriately • Implements bubble sort technique appropriately	4
Competency 10: Develops websites incorporating multi-media technologies (using HTML 5)	10.1 Explores the need for web	• The world wide web (www) • Types of web sites ○ Information, news ○ Personal, educational, commercial, Research ○ Web portals	• Describes www • Analyses the systematic arrangements of contents and structure of a web	8
	10.2 Analyses user requirements(multimedia	• Defining the objectives of a website • Contents to be displayed	• Creates effective and appropriate	4

Competency	Competency Level	Contents	Learning outcomes	Periods
	contents)		information layout of a website • Identifies the web pages of a website • Identifies the contents of a web page • Identifies navigation structure	
	10.3 Identifies appropriate HTML tags to design a single web page	• Building blocks of a web page ○ Page definition ❖ <html></html> ○ Head section ❖ <head></head> ❖ <title></title> ○ Body section ❖ <body></body> • Background color ○ Text formatting ❖ <h1>…<h6>tags ❖ ❖ ❖ Underline, bold, italic ❖ <font></font> – Size and color • Adding comments	• Analyses the arrangement of contents of a web page • Analyses the organization of contents in a web page • Creates a simple web page	4
	10.4 Uses HTML to create linked web pages	• Contents of a website ○ Home page ○ Linked pages ○ Hyperlink ❖ Different sections of the same	• Explains hypertext markup language • Identifies the standards of HTML • Saves the source	16

Competency	Competency Level	Contents	Learning outcomes	Periods
		page(book mark) ❖ Different pages of a same site(local link) ❖ Pages of different sites (External link) • Lists ○ Ordered lists ○ Unordered lists ○ Definition lists • Image • Tables ○ <table></table> ○ <th></th> ○ <tr></tr> ○ <td></td> ○ <caption> ○ Merging columns and rows • Multimedia objects ○ Audio ○ Video	document with suitable extensions • Designs the web page by inserting appropriate multimedia objects according to user requirements • Organizes data using lists and tables in the web page • Links pages and multimedia objects to the web page	
	10.5 Uses Style sheet to change the appearance of web pages	• Introduction to style sheet • CSS ○ Syntax, comments • CSS selectors ○ element, id, class, group • Ways of inserting CSS ○ Internal, external, inline • Appearance formatting ○ Background (color, image) ○ Text and fonts ○ Links	• Briefly explains style sheet and its usage • Uses the comments and correct syntax in CSS • Uses appropriate selectors to select elements in CSS • Inserts CSS in HTML web pages to	8

Competency	Competency Level	Contents	Learning outcomes	Periods
		○ Lists ○ Tables	improve the appearance • Applies various CSS formatting in HTML web pages to improve the appearance	
	10.6 Uses an authoring tool to create web pages	• Introduction to web authoring tools	• Briefly explains web authoring tools • Creates web pages using a web authoring tool	10
	10.7 Creates dynamic web pages using PHP and MySQL	• Introduction to dynamic web pages • Embedding PHP code into web page ○ Variables ○ Arrays ○ Control structures ○ Functions ○ Database connectivity ○ Working with databases • Forms ○ Input element ❖ Type attribute ❖ Name attribute ❖ Value attribute ○ Text input (Password) ○ Radio buttons ○ Check box ○ Selection	• Defines dynamic web pages • Creates data source and enters data • Creates PHP code to save/retrieve data to and from MySQL • Develop simple web based information systems	6

Competency	Competency Level	Contents	Learning outcomes	Periods
		○ Submit buttons ○ Reset button ○ Action attribute ○ Method attribute ❖ Get ❖ Post ○ Grouping form data using <fieldset> tag ○ Saving form data into database • Creating data source and entering data • Creating PHP code to retrieve data from MySQL database • Set form values using retrieved data		
	10.8 Publishes and maintains web sites	• Local publishing ○ Own computer, intranet • Internet publishing ○ Connecting to the web Service provider ○ Publishing web Pages on web server • Factors affecting performance of website	• Publishes the developed website locally • Identifies free web hosting sites from the Internet • Publishes the developed website through a free web hosting site • Investigates the factors affecting performance of website	4

Competency	Competency Level	Contents	Learning outcomes	Periods
Competency 11: Explores IoT and identify the building blocks of embedded systems to develop simple applications	11.1 Acquires the knowledge of basic building blocks of embedded systems	• Microcontroller - Based Development Systems (Arduino and other similar systems) ○ Introduction ❖ Microcontroller based Development Systems vs. microprocessor based systems ○ Features ❖ Analog Input ❖ Digital Input ❖ Microcontroller ❖ Digital Output ❖ Receiver and Transmitter Pins ❖ Communication Port ❖ Power supply ○ Connect to the computer ❖ USB Connectivity ❖ IDE Software (code editor, compiler and programmer) ○ Simple applications ❖ Switch on/off a LED ❖ Sensing ambient light intensity with a LDR and switching a LED on and off depending on the light intensity ❖ Sensing the room temperature with temperature sensor and switching a fan on and off	● Identifies and lists Microcontroller based Development Systems ● Describes available features on Microcontroller based Development Systems ● Identifies necessary software and download them from the Internet to design and write programs into Microcontroller based Development Systems ● Develops simple applications using to Microcontroller based Development Systems – Switch on/off LEDs on ambient light intensity – Run a fan on high temperature – Door open/close detection using a read switch	8

Competency	Competency Level	Contents	Learning outcomes	Periods
		depending on the temperature ❖ Detection of opening/closing a door using a read switch		
	11.2 Explores the Internet of Things (IoT) to create a simple application	● Introduction to IoT ○ Definition ○ Needs ○ IoT applications ○ Enabling technologies ● Simple IoT application to construct a remote switch	● Defines IoT (Internet of Things) ● Identifies the needs of IoT to make day to day smart ● Discusses the various applications of IoT ● Identifies the enabling technologies for IoT ● Designs and Implements an IoT application to remotely control a device through Internet Example :- ON/OFF a light ● Uses IoT based system while knowing the social	7

Competency	Competency Level	Contents	Learning outcomes	Periods
			and security consequences of IoT	
Competency 12: Explores applicability of ICT to business organizations and the competitive marketplace	12.1 Explores the role of ICT in the world of business	• Digital economy ○ New business methods in digital economy ❖ Reverse auctions ❖ Group purchasing ❖ e-Marketplace • Pure brick, brick and click, and pure click organizations • Business functions and the role of ICT ○ Accounting and ICT ○ Human resource and ICT ○ Production and ICT ○ Marketing & sales and ICT ○ Supply chain management and ICT ○ Business communication and ICT ○ Secure payment mechanisms ❖ Payment gateways ❖ Secure credit card payments ❖ Third party systems – PayPal etc ❖ Mechanisms	• Defines digital economy • Lists and describes new business methods in digital economy • Identifies the concepts behind pure brick, brick and click, and pure click organizations • Describes the role of ICT in business functions of an organization	4

Competency	Competency Level	Contents	Learning outcomes	Periods
		– Data encryption – Micro credit payments (bit coin etc.) • Threats and opportunities in ecommerce ○ Privacy ○ Product commercialization		
	12.2 Analyses the relationship between ICT and business operations	• E-Commerce and e-business ○ The scope of e-commerce and e-business ○ Types of e-business transactions ❖ B2B, B2C, C2C, C2B, B2E, G2C • E-Business ○ Virtual storefronts ○ Information brokers ○ Online marketplace ○ Content provider ○ Online service provider ○ Portal ○ Virtual Community • Advantages and disadvantages of e-business	• Distinguishes the e-commerce and e-business • Investigates the scope of e - commerce and e-business • Lists and briefly describes the types of e-business transactions • Identifies the advantages and disadvantages of e-business	4
	12.3 Analyses the ICT in terms of generating and delivering an improved products and services to consumers	• E-marketing ○ Concepts of marketing ○ Use of ICT in marketing ❖ Web advertising etc • Databases in marketing ○ Predicting customer behavior with AI tools and techniques	• Defines e-marketing • Identifies the role of ICT in e-marketing • Investigates the usage of database in marketing	4

Competency	Competency Level	Contents	Learning outcomes	Periods
		○ Gaining competitive advantages through ICT • Mobile Marketing	activities to improve the product and services according to the requirements of the customers • Identifies the ways of gaining competitive advantages using ICT	
Competency 13: Explores new trends and future directions of ICT	13.1 Explores new trends and future directions in computing	• Intelligent and emotional computing • Artificial intelligence • Man-machine coexistence • Machine to machine coexistence	• Describes intelligent and emotional computing. • Explains artificial intelligences • Appreciates man-machine coexistences	4
	13.2 Explores the fundamentals and applications of agent technology	• Software agents • Multi-agent systems • Applications of Agent systems	• Briefly describes software agents and their characteristics • Briefly describes multi-agent systems and their characteristics • Identifies the applications of agent systems	4